

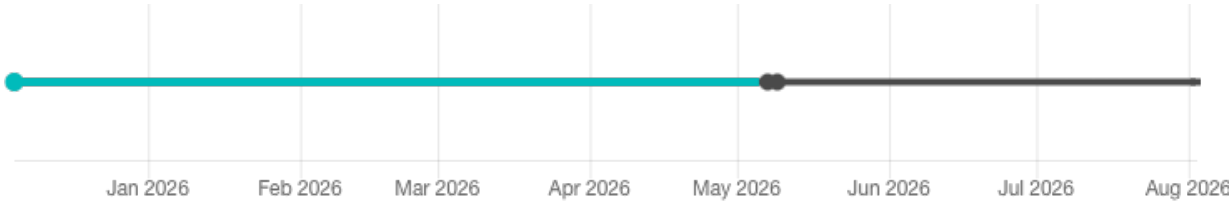


AGRIPOTENTIAL: VITICULTURE EDITION

20 PARTICIPANTS

185 SUBMISSIONS

ORGANIZED BY: Melsakka
CURRENT PHASE ENDS: May 7, 2026 At 5:30 AM
GMT+5:30
CURRENT SERVER TIME: May 4, 2026 At 10:33 AM
GMT+5:30
Docker image: [codalab/codalab-legacy:py37](#) 



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Competition Overview

Welcome to the AgriPotential Viticulture Challenge 🍷! This competition tasks participants with developing machine learning models to accurately predict the agricultural potential of land specifically for viticulture (grapevine cultivation) using high-resolution, multi-temporal remote sensing data.

The challenge uses a subset of the novel AgriPotential dataset, focusing exclusively on the viticulture label. Your goal is to map the capacity of different land parcels to support vine growth

based on satellite imagery captured over time.

The Problem

Agricultural potential is defined by a set of environmental factors (soil, climate, topography) that determine the suitability for a specific crop. Using multi-spectral and multi-temporal Sentinel-2 satellite imagery from 2017 to 2019, you must build a model capable of predicting the five-class ordinal potential of each land pixel for viticulture.

Objective

Develop a robust model that can predict the ordinal class of viticulture potential for each pixel:

- Very Low
 - Low
 - Average
 - High
 - Very High
-

Dataset Details

The competition dataset is derived from the AgriPotential benchmark, covering diverse agricultural landscapes in Southern France. The first original dataset was first introduced in the [AgriPotential Paper](#). For this challenge we will be using another version of the data that is described below.

Input Data

34 Sentinel-2 images collected between 2017 and 2019. Each image has 10 spectral bands (more details in appendix). The data is acquired at irregular intervals in time.

Target Label (Ground Truth)

Pixel-level annotations representing the viticulture potential. Each pixel has 1 of 5 possible values:

- Very Low
- Low
- Average
- High
- Very High

Data Access

A tutorial on how to access the data is presented in a Google Colab notebook. PyTorch Dataset and DataLoader objects are provided but the users are free to implement their own dataloaders. URL: https://colab.research.google.com/drive/1Ys_z7qBY1pa8iY7PMaLzDsTuiFaJDhHu

Key Difficulties

Spatio-Temporal Modeling

Spatio-temporal modeling is a powerful analytical approach that explicitly incorporates both the

spatial dimension (location) and the temporal dimension (time) into a unified framework to understand, describe, and predict phenomena that vary across space and evolve over time.

In the context of remote sensing, this means moving beyond analyzing a single image or a sequence of images independently. Instead, it involves leveraging the full information contained in a Satellite Image Time Series (SITS), which are multiple images of the same area taken over different dates, where each pixel's value is influenced by both its geographical surroundings and its history. More details about STIS in the Appendix.

Spectral Modeling

Spectral modeling is a core technique of remote sensing that analyzes information which is invisible to the naked eye. The spectral dimension refers to the measurement of the reflected sunlight across the electromagnetic spectrum. Modeling the complex interaction and evolution of electromagnetic wavelengths in time could be a key component in predicting agricultural potentials.

Ordinal Classification

Ordinal Classification (sometimes referred to as Ordinal Regression) is a type of classification where the categories have a meaningful, inherent order or rank, but the distance between the categories is either unknown or unequal. Designing a model to respect this ordinal relationship is crucial when misclassifications closer to the true class are less problematic than misclassifications farther away. In this challenge, the five potential classes have a clear rank:

$\text{Very Low} < \text{Low} < \text{Medium} < \text{High} < \text{Very High}$

The core challenge is ensuring the model's objective function penalizes large errors (e.g., predicting "Very Low" when the truth is "Very High") more severely than small errors (e.g., predicting "High" when the truth is "Very High").

! Note

The preceding points are provided for context and guidance only. Using them is optional and not mandatory.

Submission

The required submission is a single ZIP archive containing the segmentation results for the test subset (test.csv). Inside this archive, you must include the segmented images as PNG files. Crucially, each PNG file must be named exactly after its corresponding patch_id. The goal is to submit all 800 images; while submitting fewer is permissible, it will negatively impact your overall score, and any extraneous files beyond the expected PNGs will be ignored by the scoring system. Additionally, the predicted output must only contain integer values ranging from 0 (very low) to 4 (very high).

ALL FILES SHOULD BE ROOT OF THE ZIP (no subfolders)

Note that the true labels for the test data are hidden from participants and are retained by the organizers for evaluation.

We also **strongly** encourage participants to submit a PDF file explaining their method inside of the zip. In that case the name of the pdf should **strictly** be "report.pdf" otherwise it will be ignored.

The submission of the report is *optional*, unless otherwise is required (see Phases and News)

Example (Pseudo code):

```
input_loader = get_input_loader(subset="test")
model = MyModel()
for data, patch_id in input_loader:
    output = model(data)
    save_file(output, patch_id+".png")
make_zip()
```

Evaluation

Agricultural potentials are represented by ordered classes from “very low” to “very high”. The evaluation will be conducted using the Accuracy ± 1 metric, known as “Accuracy with ± 1 tolerance,” which measures the proportion of predictions that are within one class of the true label. The ordinal nature of the classes allows predictions that are close to the true class to be considered partially correct. The final metric is the Overall Accuracy calculated across all pixels. The formula of the used accuracy is given below:

$$Accuracy_{\pm 1} = \frac{1}{N} \sum_{i=1}^N 1 \left(|y_i^{\text{true}} - y_i^{\text{pred}}| \leq 1 \right)$$

Example:

```
# Simplified evaluation for illustration purpose only.
# model's outputs is a segmentation map
predictions = [
    [1, 2, 3],
    [2 0, 4],
    [1, 4, 4],
]

truth = [
    [1, 2, 4],
    [2 1, 2],
    [1, 4, 4],
]

accuracy_pm_1 = compute_acc() # -> 88.89% accuracy
```

Citation

If you find this work helpful please cite the following:

```
@article{sakka2025agripotential,  
title={AgriPotential: A Novel Multi-Spectral and Multi-Temporal Remote Sensing Dat  
author={Sakka, Mohammad El and De Pourtales, Caroline and Chaari, Lotfi and Mothe,  
journal={arXiv preprint arXiv:2506.11740},  
year={2025}  
}
```

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v1.26